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InterpN: Fast Interpolation

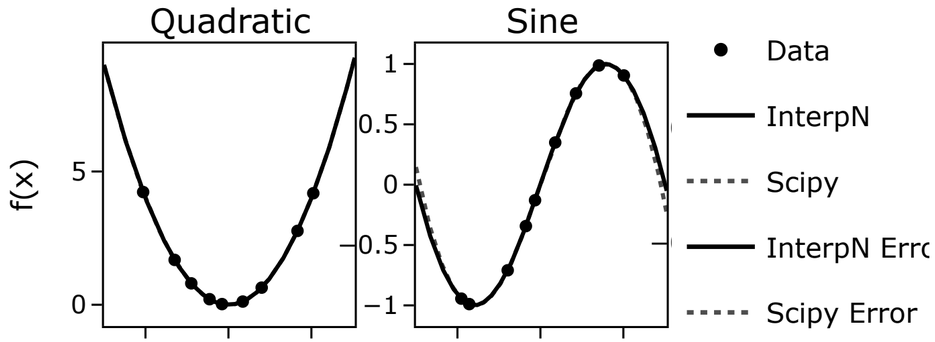
2026-Jan-22

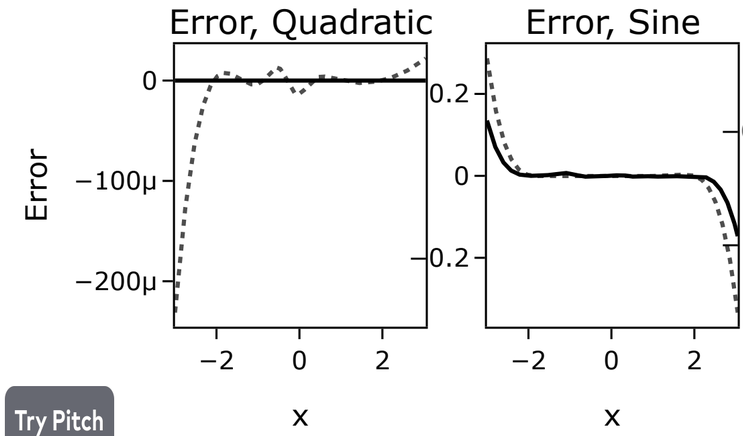
Contents

Motivation & Results 1/4

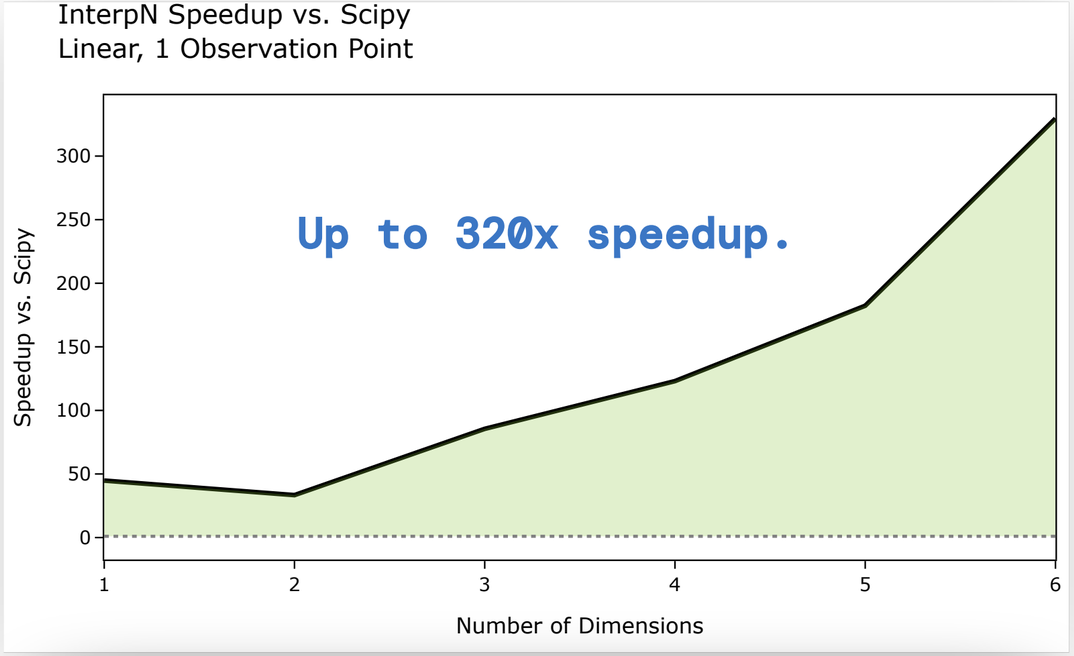
We need to estimate the values between data points

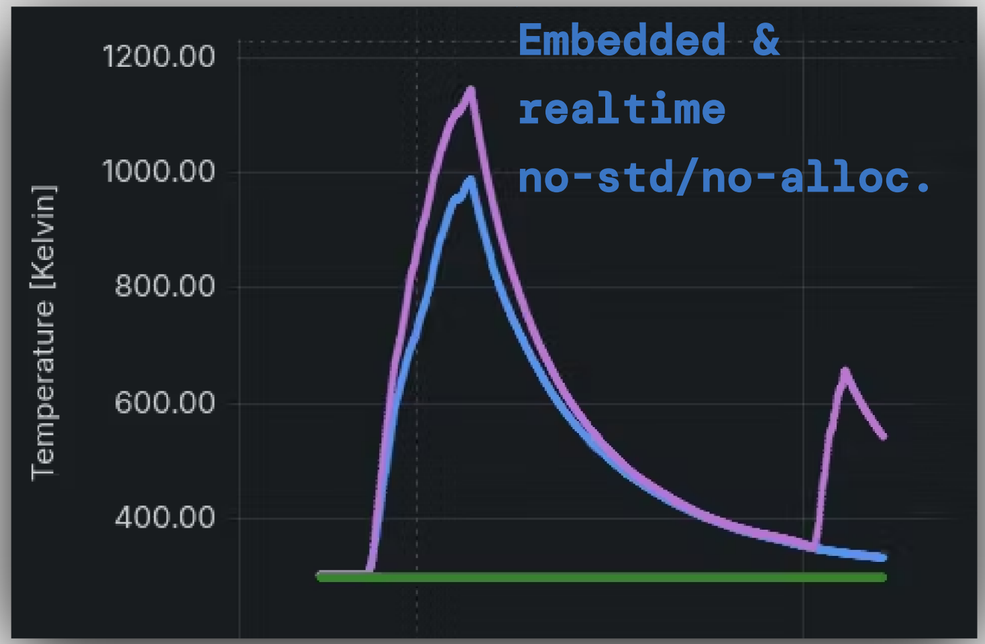
- Sometimes data points are expensive
- Sometimes data points have analytic significance
- InterpN supports both embedded *and* scientific computing





About 8 orders of magnitude less float error.





Try Pitch

Rust before Code

Want your program to be fast? Just ask the compiler.

Cube → Tree → Loop

The Algorithm

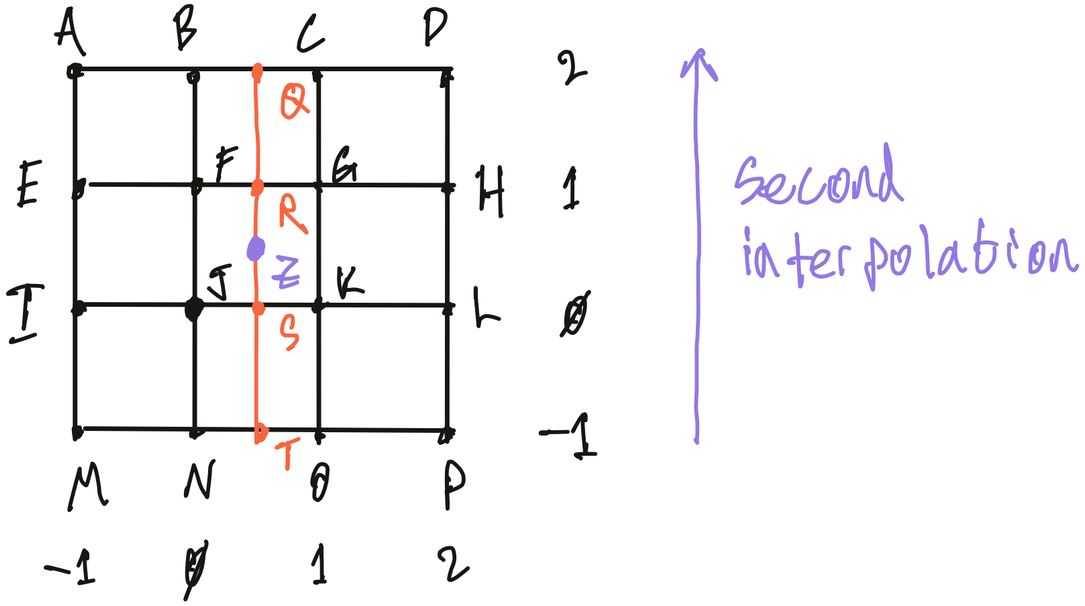
An N-cube around the sample contributes to its value

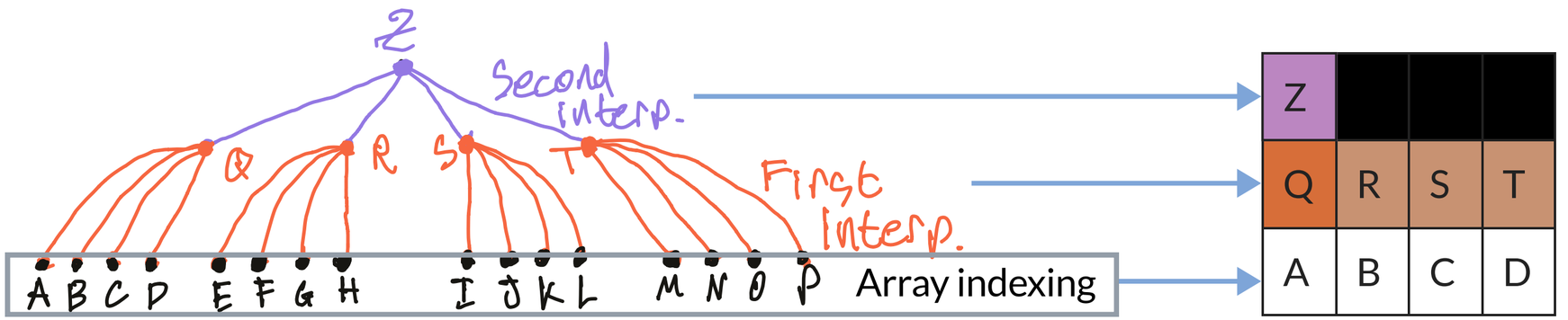
- X^N can be a lot of data
- Don't want to allocate on a microcontroller

Solution: fixed-size storage on the stack

- No allocation; linear storage instead of exponential

Don't allocate until you have exhausted all other options!





Project Configuration

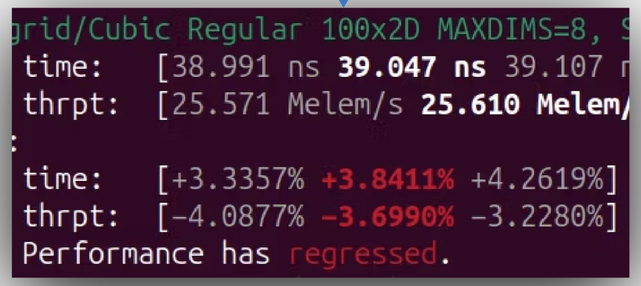
Cargo.toml

Release Profile

```
[profile.release]
opt-level = 3
codegen-units = 1
lto = true
overflow-checks = true
```

More important for larger projects.

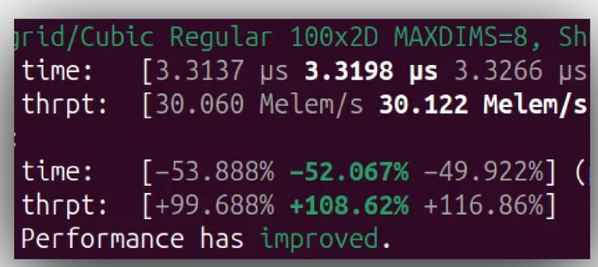
```
# Even for embedded!
# Optimize everything at once
# Clean up the binary
# Free(ish) reliability!
```



.cargo/config.toml

x86 Instruction Sets

```
[target.'cfg(any(target_arch = "x86_64", target_arch = "x64"))']
rustflags = ["-C", "target-cpu=x86-64-v3"]
# When building for 64-bit x86 targets, use a reference CPU.
# See: “Microarchitecture Levels” on wikipedia
#
#
# x86-64-v1 → 2003
# x86-64-v2 → 2008 (SSE4)
# x86-64-v3 → 2013 (AVX2, FMA)
# x86-64-v4 → 2017 (danger! high latency AVX512)
```

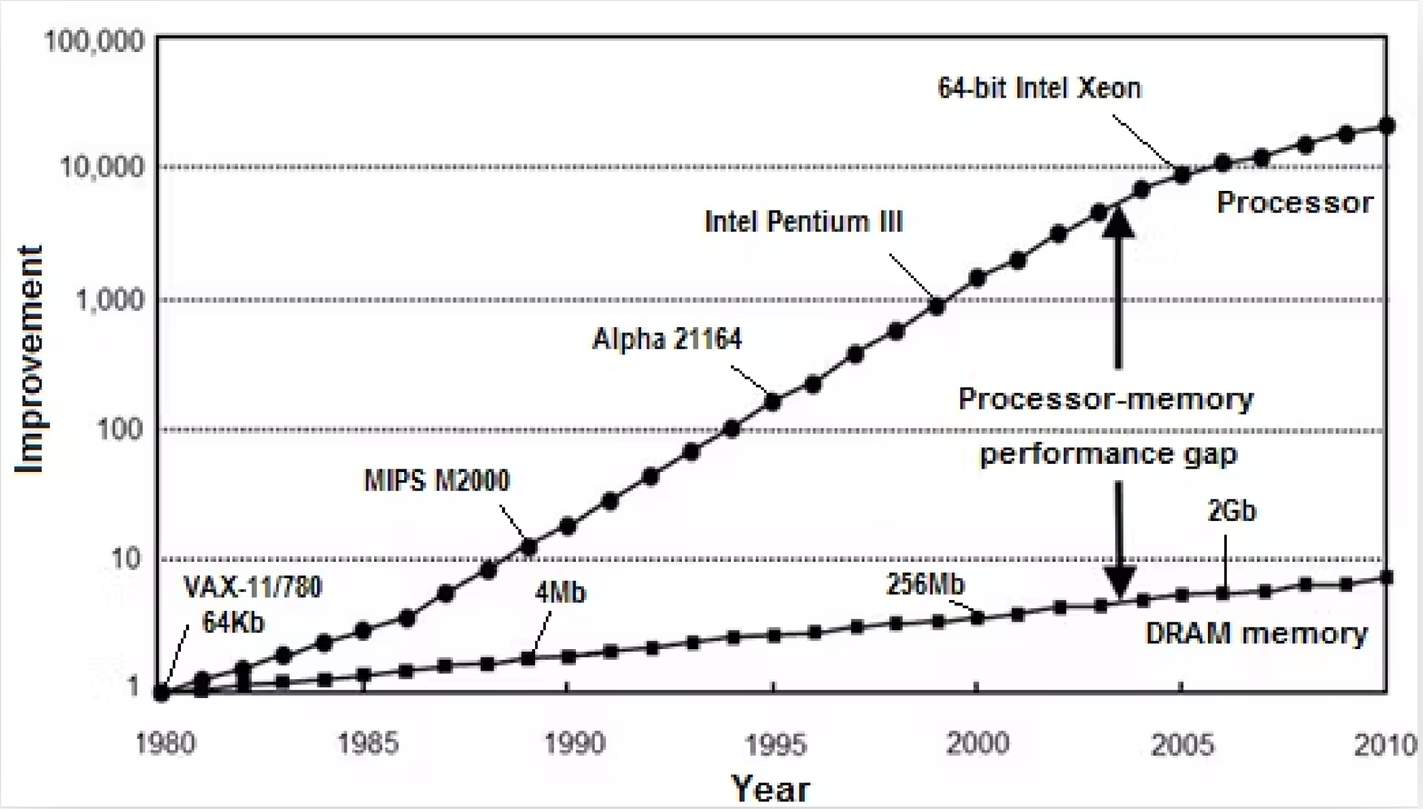


Vectorization

We've been holding it sideways.

The CPU-RAM gap is not an abstract idea

Vectorization 1/6



Things have changed!

RAM is extremely bottlenecked now.

Computing norms were established in the 1990s!

Optimize to reduce RAM traffic over CPU cycles.

D. Efnusheva, A. Cholakoska, and A. Tentov, "A Survey of Different Approaches for Overcoming the Processor - Memory Bottleneck," International Journal of Information Technology and Computer Science, vol. 9, p. 151, Apr. 2017, doi: 10.5121/ijcsit.2017.9214.

Parallelism without threads. It's free compute

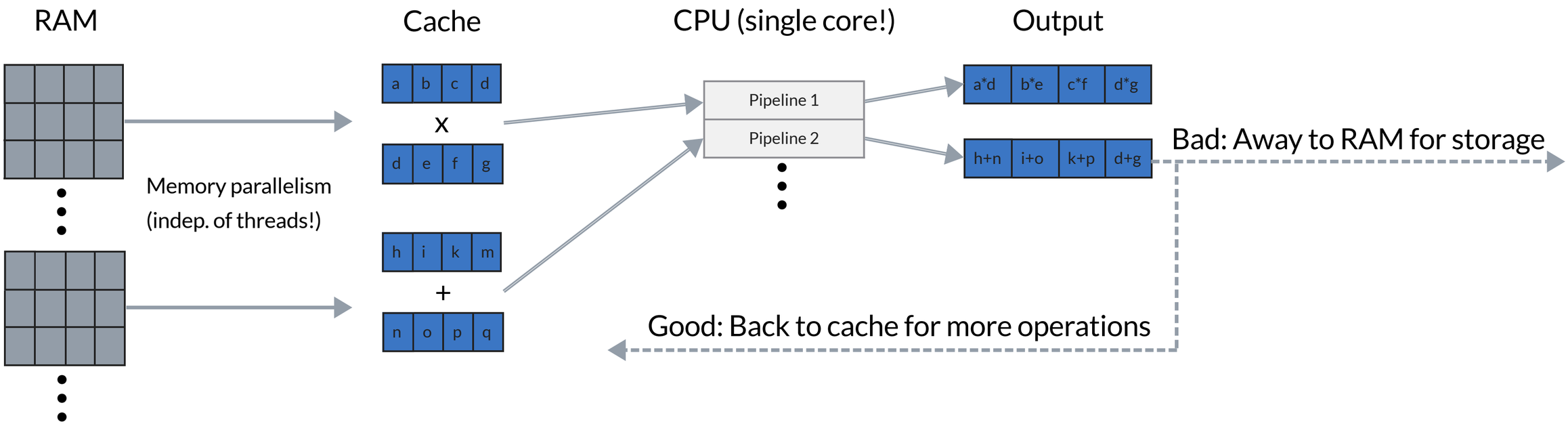
Vectorization 2/6

Vectorization

- SIMD / Single Instruction Multiple Data
- Disrupted by branches (including panic branches!)
- Show your code to the compiler!

Pipelining

- CPU can do multiple independent things at once
- Distinct from threading
- Show your code to the CPU!



Vectorization 3/6

Type 1: Simple Loops

⚠️ This produces slow traffic back-and-forth to RAM.

⚠️ Extra allocations increase latency and total RAM usage.

⚠️ The compiler and CPU can only optimize one operation at a time.

⚠️ Bounds checks are repeated for each operation.

✅ Formula does not have to be compiled together.

```rust
fn mul(a: &[f64], b: &[f64]) -> Result<Vec<f64>, &'static str> {
    // Check bounds before loop!
    // Otherwise, it will not vectorize
    let n: usize = a.len();
    if b.len() != n {
        return Err("Dimension mismatch");
    }

    // Allocate storage
    let mut out: Vec<f64> = vec![0.0; n];

    // Do the calculations
    for i: usize in 0..n {
        out[i] = a[i] * b[i];
    }

    Ok(out)
}
```

Vectorization 4/6

Type 2: Complex Loops

✅ Minimum possible RAM accesses and heap allocations.

✅ Compiler automatically reuses duplicate values.

✅ CPU can execute parallel operations without threading.

✅ Bounds checks are only performed once.

✅ Compiler generates vector instructions.

✅ Branches are okay!

✅ Linear speedup when wrapped with threading.

⚠️ Formula must be compiled together.

Inline everything

```rust
#[inline]
fn sub(a: [f64; 3], b: [f64; 3]) -> [f64; 3] {
    [a[0] - b[0], a[1] - b[1], a[2] - b[2]]
}
```

```rust
#[inline]
fn dot(u: [f64; 3], v: [f64; 3]) -> f64 {
    u[0] * v[0] + u[1] * v[1] + u[2] * v[2]
}
```

```rust
#[inline]
fn cross(u: [f64; 3], v: [f64; 3]) -> [f64; 3] {
    [
        u[1] * v[2] - u[2] * v[1],
        u[2] * v[0] - u[0] * v[2],
        u[0] * v[1] - u[1] * v[0],
    ]
}
```

```rust
#[inline]
fn norm(u: [f64; 3]) -> f64 {
    dot(u, u).sqrt()
}
```

scalar formula

```rust
/// tetrahedron with the first vertex as the origin.
/// https://en.wikipedia.org/wiki/Solid_angle#Tetrahedron
#[inline]
pub fn solid_angle_tetrahedron_scalar(
    v0: [f64; 3],
    v1: [f64; 3],
    v2: [f64; 3],
    v3: [f64; 3],
) -> f64 {
    // Vertex vectors
    let (a, b, c) = (sub(v1, v0), sub(v2, v0), sub(v3, v0));  // (m)
    let (la, lb, lc) = (norm(a), norm(b), norm(c)); // (m) Vertex vector lengths
    let abc = la * lb * lc;  // (m^3) Length product. Branch determined here!

    // Solid angle
    let triple = dot(a, cross(b, c)); // (m^3) Scalar triple product
    let denom = abc + dot(a, b) * lc + dot(a, c) * lb + dot(b, c) * la;  // (m^3)
    let angle = 2.0 * libm::atan2(triple, denom); // (rad) f64::atan2 defers to libc

    // Check for degeneracy _last_ to avoid disrupting flow
    if abc != 0.0 {
        angle
    } else {
        0.0
    }
}
```

vector formula

```rust
/// Vector variant of [solid_angle_tetrahedron_scalar]
#[inline] // Enable cross-crate inlining
pub fn solid_angle_tetrahedron(
    tetrahedra: &[[[f64; 3]; 4]],
    out: &mut [f64],
) -> Result<(), &'static str> {
    // Check bounds
    let n = out.len();
    if tetrahedra.len() != n {
        return Err("Dimension mismatch");
    }

    // Do calculations
    for i in 0..n {
        let tet = tetrahedra[i];
        out[i] = solid_angle_tetrahedron_scalar(tet[0], tet[1], tet[2], tet[3]);
    }

    Ok(())
}
```

How do we know if it worked?

Vectorization 5/6

Benchmarks

Mess it up and see if it gets worse!

Fix it and see if it gets better!

`**criterion**` is great for this.

bounds checks!!

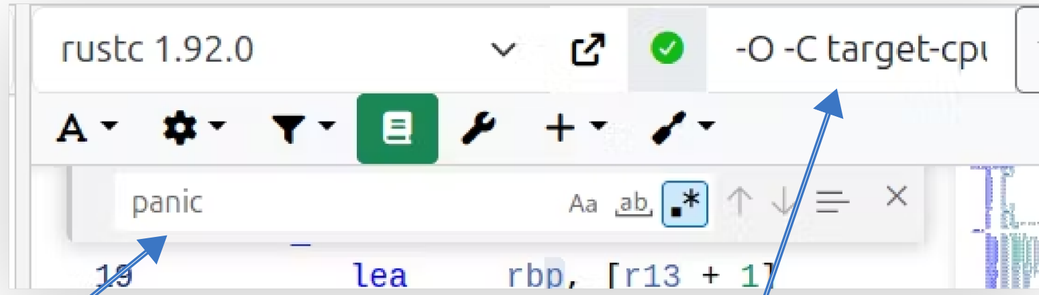
```
led_100-grid/Linear Regular 100x1D MAXDIMS=1, Shuffled Order/1
        time:   [7.5980 ns 7.6594 ns 7.7262 ns]
        thrpt:  [129.43 Melem/s 130.56 Melem/s 131.61 Melem/s]
 change:
        time:   [+35.729% +37.337% +38.877%] (p = 0.00 < 0.05)
        thrpt:  [-27.994% -27.186% -26.324%]
        Performance has regressed.
```

```
ed_100-grid/Linear Regular 100x1D MAXDIMS=1, Shuffled Order/1
        time:   [5.6419 ns 5.6935 ns 5.7460 ns]
        thrpt:  [174.03 Melem/s 175.64 Melem/s 177.24 Melem/s]
 change:
        time:   [-26.692% -25.793% -24.916%] (p = 0.00 < 0.05)
        thrpt:  [+33.184% +34.757% +36.411%]
        Performance has improved.
```

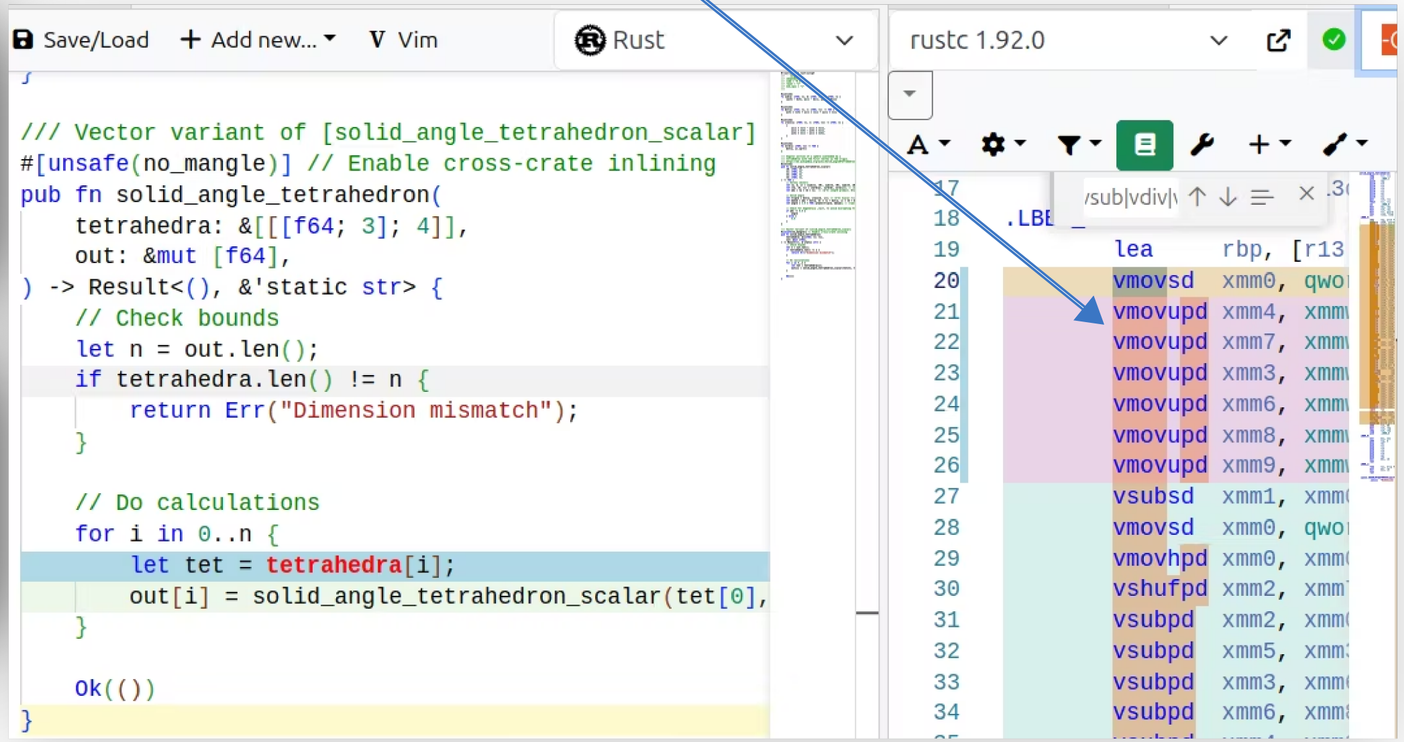
Assembly

Go inside and look!

- Single file: use Godbolt!
- Multi-file: use cargo-show-asm!
- Look for `panic` and packed ops.



optimizations!!



Try Pitch

A word about allocation

Vectorization 6/6

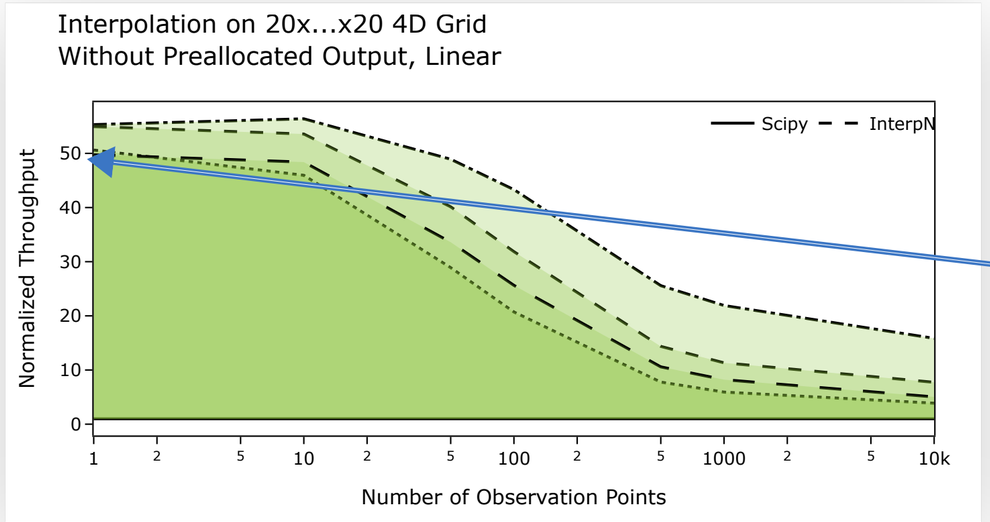
Allocation is the enemy

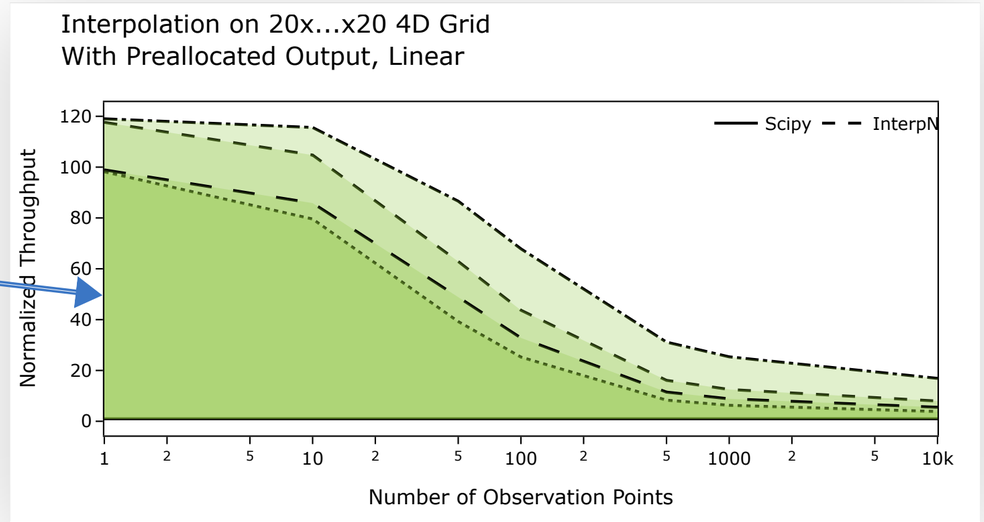
- Effect depends on exact use pattern; **masked by benchmarking.**
- Inter-thread and inter-process interference & extra RAM accesses.

At least give users the option to pre-allocate output!

2x speedup for small sizes.

Just stop allocating!





Telling rustc what you know about the program

Compile-Time Loop Unrolling 1/2

- Problem: house is too cold
- Solution: unroll large loops with `**crunchy**`

This can be a const generic bound!

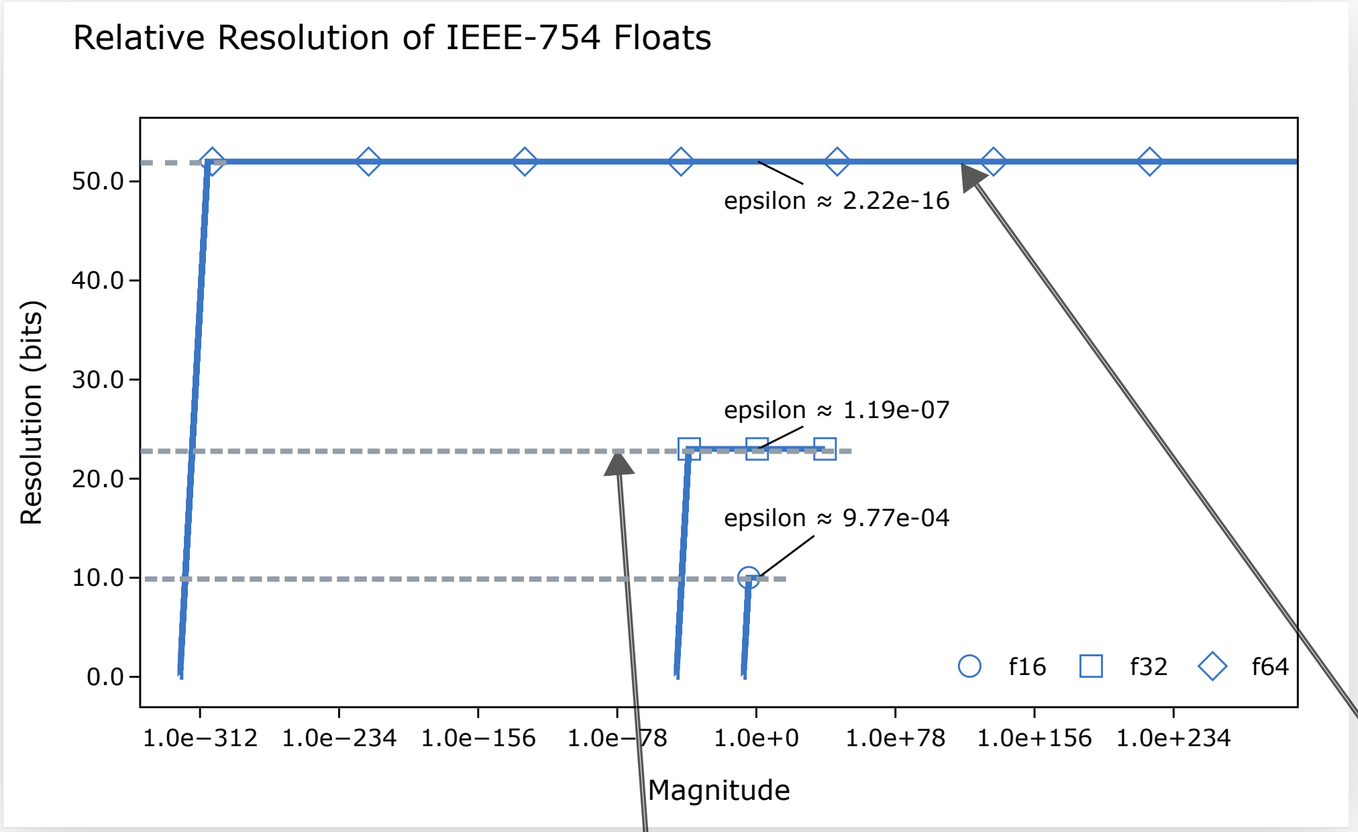
```
if N <= 3 {
    unroll! {
        for i < 64 in 0..nverts {  // const loop
            unroll_vertices_body!(i);
        }
    }
} else {
    for i: usize in 0..nverts {
        unroll_vertices_body!(i);
    }
}
```

```
rid/Cubic Regular 100x2D MAXDIMS=8, Sh
time:   [3.1936 µs 3.1989 µs 3.2054 µs
thrpt:  [31.197 Melem/s 31.260 Melem/s

time:   [-31.332% -30.524% -29.658%] (
thrpt:  [+42.163% +43.934% +45.628%]
Performance has improved.
```

Two flops for the price of one (ish)

Fused Multiply-Add (FMA) 1/3



Good for application code, not for embedded

- `a.mul_add(x, b)` → `a*x + b` in a single step.
- Less roundoff error. Float resolution is real!
- Some subtlety; still worth it.

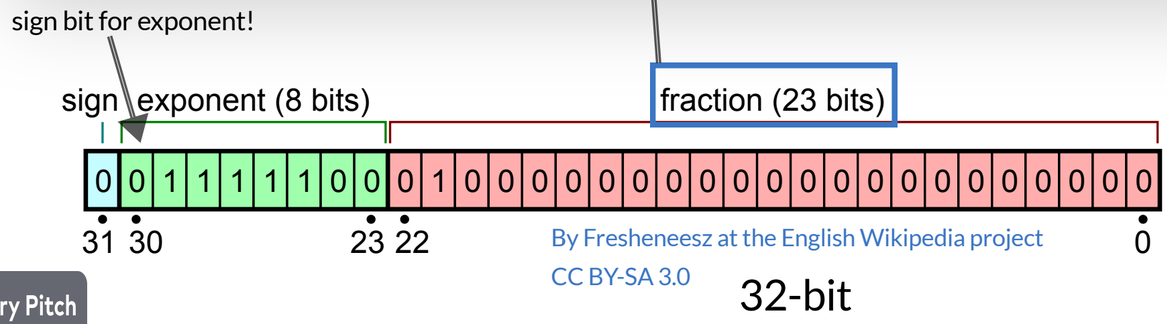
Demystify floats

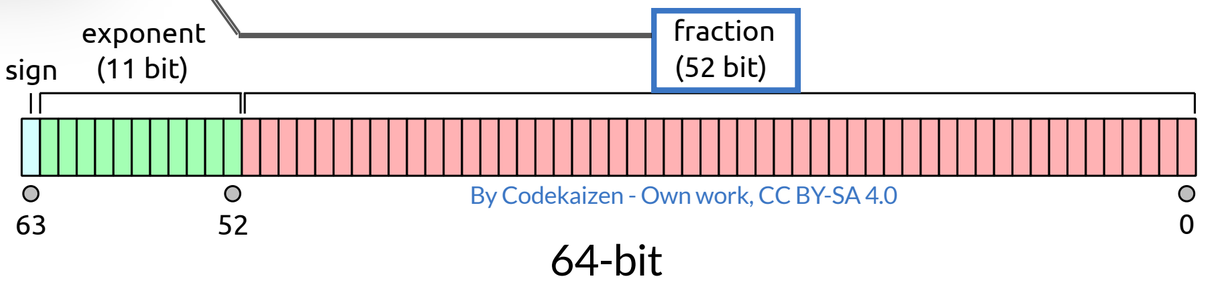
- It's just a (near-)constant relative resolution!
- You can do error accounting in your head!
- Do not abide a -ffast-math

$$e = 0 \Rightarrow x = (-1)^s \left(\sum_{i=1}^{52} b_i 2^{-i} \right) 2^{-1022}$$

$$1 \le e \le 2046 \Rightarrow x = (-1)^s \left(1 + \sum_{i=1}^{52} b_i 2^{-i} \right) 2^{e-1023}$$

$$e = 2047 \Rightarrow x = \pm\infty \text{ or NaN}$$





The return of the solid angle

Fused Multiply-Add (FMA) 2/3

```rust
#[inline]
fn dot(u: [f64; 3], v: [f64; 3]) -> f64 {
    u[0] * v[0] + u[1] * v[1] + u[2] * v[2]
}

#[inline]
fn cross(u: [f64; 3], v: [f64; 3]) -> [f64; 3] {
    [
        u[1] * v[2] - u[2] * v[1],
        u[2] * v[0] - u[0] * v[2],
        u[0] * v[1] - u[1] * v[0],
    ]
}

#[inline]
fn norm(u: [f64; 3]) -> f64 {
    dot(u, u).sqrt()
}

/// Angular portion of a sphere subtended by a
/// tetrahedron with the first vertex as the orig
/// https://en.wikipedia.org/wiki/Solid_angle#Tet
#[inline]
pub fn solid_angle_tetrahedron_scalar(
    v0: [f64; 3],
    v1: [f64; 3],
    v2: [f64; 3],
    v3: [f64; 3],
) -> f64 {
    // Vertex vectors
    let (a, b, c) = (sub(v1, v0), sub(v2, v0), su
    let (la, lb, lc) = (norm(a), norm(b), norm(c)
    let abc = la * lb * lc;  // (m^3) Length prod

    // Solid angle
    let triple = dot(a, cross(b, c)); // (m^3) Sc
    let denom = abc + dot(a, b) * lc + dot(a, c)
    let angle = 2.0 * libm::atan2(triple, denom);

    // Check for degeneracy _last_ to avoid disru
    if abc != 0.0 {
        angle
    } else {
        0.0
    }
```

No FMA

```rust
#[inline]
fn dot(u: [f64; 3], v: [f64; 3]) -> f64 {
    u[0].mul_add(v[0], u[1].mul_add(v[1], u[2] * v[2]))
}

#[inline]
fn cross(u: [f64; 3], v: [f64; 3]) -> [f64; 3] {
    [
        u[1].mul_add(v[2], -u[2] * v[1]),
        u[2].mul_add(v[0], -u[0] * v[2]),
        u[0].mul_add(v[1], -u[1] * v[0]),
    ]
}

#[inline]
fn norm(u: [f64; 3]) -> f64 {
    dot(u, u).sqrt()
}

/// Angular portion of a sphere subtended by a
/// tetrahedron with the first vertex as the origin.
/// https://en.wikipedia.org/wiki/Solid_angle#Tetrahedron
#[inline]
pub fn solid_angle_tetrahedron_scalar(
    v0: [f64; 3],
    v1: [f64; 3],
    v2: [f64; 3],
    v3: [f64; 3],
) -> f64 {
    // Vertex vectors
    let (a, b, c) = (sub(v1, v0), sub(v2, v0), sub(v3, v0)); // (m)
    let (la, lb, lc) = (norm(a), norm(b), norm(c)); // (m) Vertex vector lengths
    let abc = la * lb * lc; // (m^3) Length product. Branch determined here!

    // Solid angle
    let triple = dot(a, cross(b, c)); // (m^3) Scalar triple product
    let denom = dot(a, b).mul_add(lc, dot(a, c).mul_add(lb, dot(b, c).mul_add(la, abc))); // (m^3)
    let angle = 2.0 * libm::atan2(triple, denom); // (rad) f64::atan2 defers to libc

    // Check for degeneracy _last_ to avoid disrupting flow
    if abc != 0.0 {
        angle
    } else {
        0.0
    }
```

Yes FMA

rustc 1.92.0 -O -C target-cpu=x86-64-v3

vfm(add|sub) 1 of 18

```
38
39      vmulpd  xmm0, xmm8, xmm8
40      vfmadd231pd     xmm0, xmm6, xmm6
41      vfmadd231pd     xmm0, xmm4, xmm4
42      vsqrtpd xmm5, xmm0
43      vmulsd  xmm0, xmm1, xmm1
44      vfmadd231sd
45      vfmadd231sd
46      vsqrtsd xmm7,
47      vshufpd xmm9, xmm5, xmm5, 1
48      vmulsd  xmm0, xmm9, xmm5
49      vmulsd  xmm15, xmm0, xmm7
50      vmovsd  qword ptr [rsp], xmm15
51      vshufpd xmm10, xmm8, xmm8, 1
52      vmulsd  xmm11, xmm10, xmm3
53      vshufpd xmm12, xmm6, xmm6, 1
54      vfmsub231sd     xmm11, xmm12, xmm1
55      vshufpd xmm13, xmm4, xmm4, 1
56      vmulsd  xmm14, xmm13, xmm1
57      vfmsub231sd     xmm14, xmm10, xmm2
58      vmulsd  xmm0, xmm12, xmm2
59      vfmsub231sd     xmm0, xmm13, xmm3
60      vmulsd  xmm0, xmm8, xmm0
61      vfmadd231sd     xmm0, xmm6, xmm14
62      vfmadd231sd     xmm0, xmm4, xmm11
63      vmulsd  xmm11, xmm8, xmm10
64      vfmadd231sd     xmm11, xmm6, xmm12
65      vfmadd231sd     xmm11, xmm4, xmm13
66      vmulsd  xmm8, xmm8, xmm1
67      vfmadd231sd     xmm8, xmm3, xmm6
68      vfmadd231sd     xmm8, xmm2, xmm4
69      vmulsd  xmm1, xmm10, xmm1
70      vfmadd231sd     xmm1, xmm12, xmm3
71      vfmadd231sd     xmm1, xmm13, xmm2
72      vfmadd213sd     xmm1, xmm5, xmm15
73      vfmadd231sd     xmm1, xmm9, xmm8
74      vfmadd231sd     xmm1, xmm7, xmm11
75      call    r12
```

18% Vector FMA

Try Pitch

Fused Multiply-Add (FMA) 3/3

Cargo.toml

Add an `fma` feature

```
[features]
default = []        # Not on by default!
fma = []
python = [“fma”]# Enable for python
```

Code

Use FMA if enabled

```rust
// Horner's method
#[cfg(not(feature = "fma"))]
{
    y0 + t * (c1 + t * (c2 + t * c3))
}
#[cfg(feature = "fma")]
{
    c3.mul_add(t, c2).mul_add(t, c1).mul_add(t, y0)
}
```

Perf

Profit

```
grid/Linear Regular 100x1D MAXDIMS=1,
 time:   [320.70 ns 321.86 ns 323.31
 thrpt:  [309.30 Melem/s 310.69 Melem
:
 time:   [-13.441% -12.630% -11.896%]
 thrpt:  [+13.502% +14.456% +15.528%]
Performance has improved.
```

Rust after Code

The myth of premature optimization.

Telling rustc what you know about the user

Profile-Guided Optimization (PGO) for Python Extensions 1/3

Python extensions are good for adoption & distribution

- 10k downloads/month vs. 20k all-time, in this case

General guidelines for Python Extensions

- Use pyo3+maturin with ABI3; target 2013+ (x86-64-v3)
- Publish as both a Rust crate *and* a Python library
- `maturin generate-ci`
- Document units-of-measure & cite sources!!
- Do PGO if you can phrase a workload

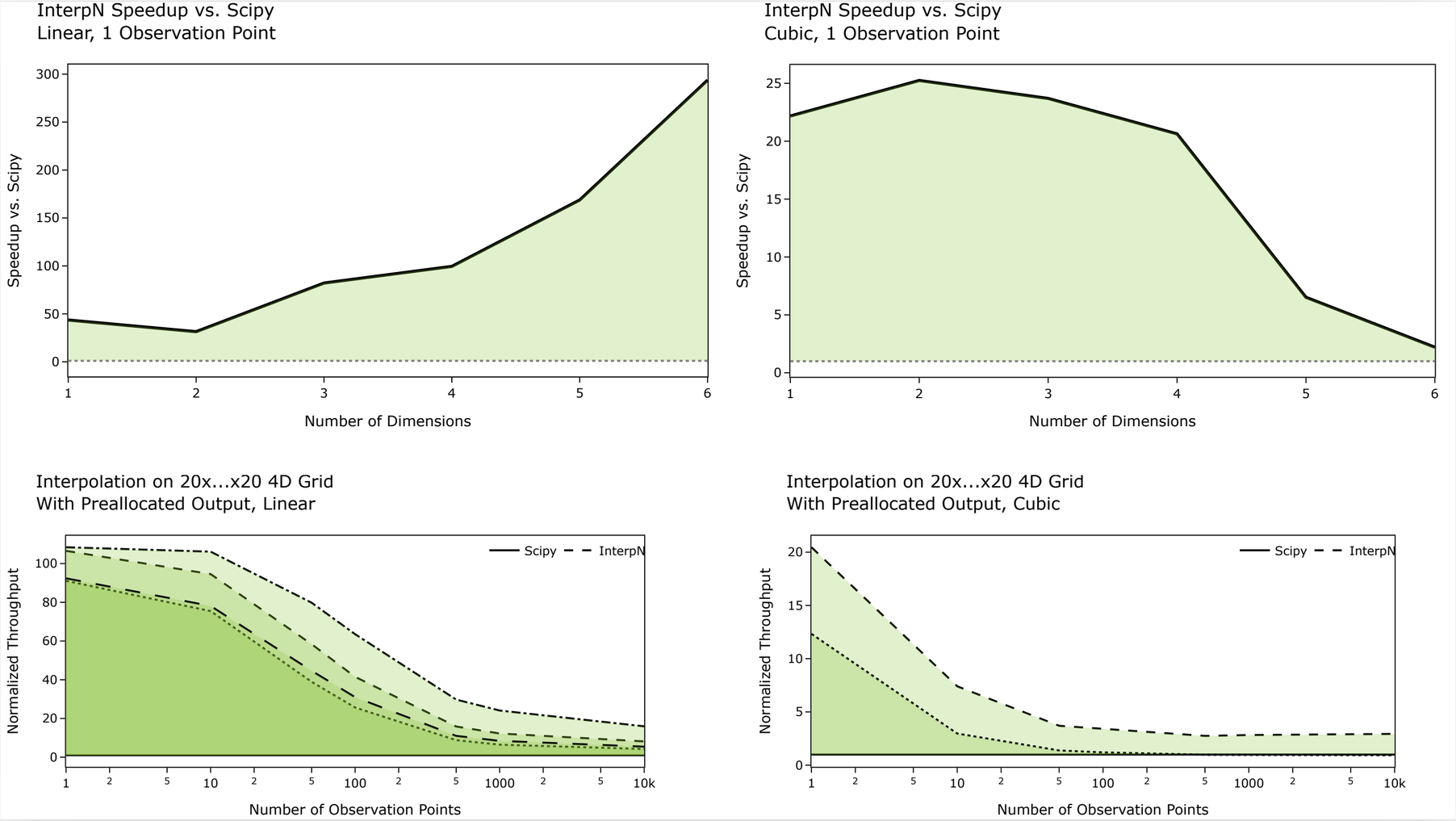
PGO Procedure

1. Build instrumented (just extra rustc args via maturin!)
2. Run PGO workload (just a python script!)
3. Process profile outputs (just call llvm on some files!)
4. Build optimized (just extra rustc args via maturin!)

Bottom line: 15-30% speedup for most-used cases.

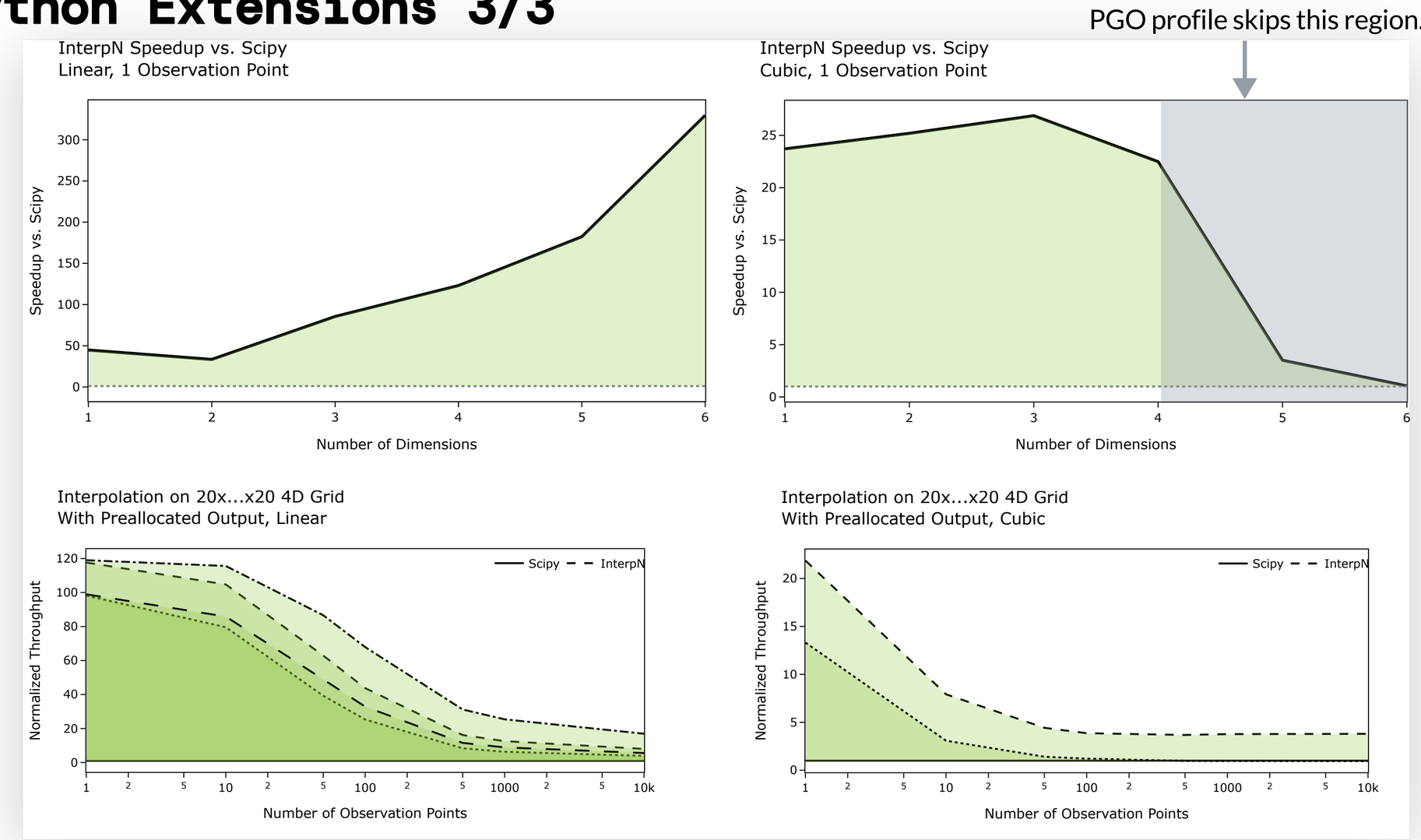
Before

Profile-Guided Optimization (PGO) for Python Extensions 2/3



After

Profile-Guided Optimization (PGO) for Python Extensions 3/3

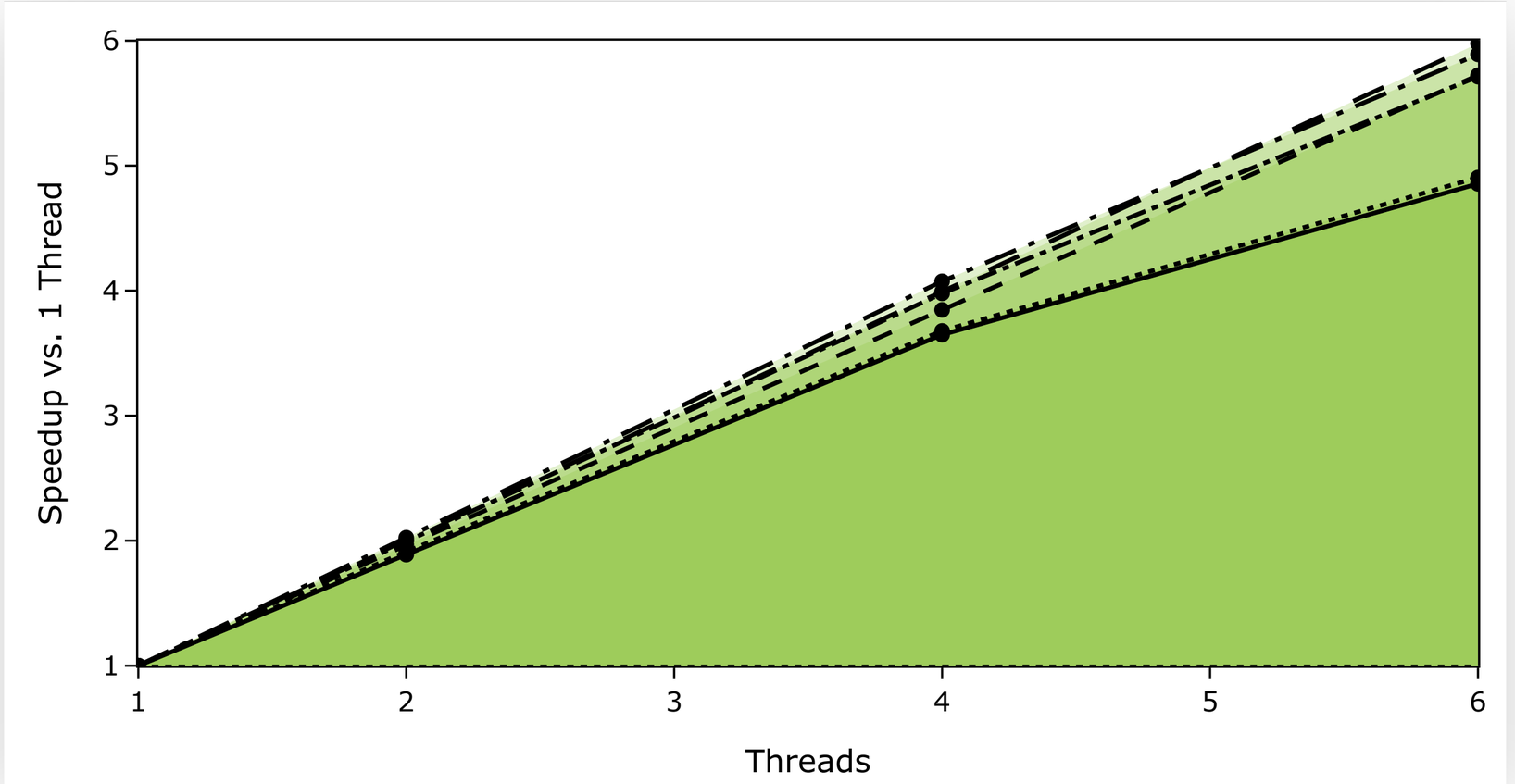


Do this *last*, not first!

Thread Parallelism 1/3

Near-linear speedup; limited only by thread spawning overhead & CPU heating

- Chunk slices for each thread → no allocation, message-passing, or locks
- Use global lazy static to check number of physical cores once and never again
- Only thread if it's faster than single-core
- Don't voluntarily yield the thread



Benchmark case representative of a 10-megapixel image.

Recipes, or it didn't happen

Thread Parallelism 2/3

✅ No allocation

- Slice reference inputs
- Stack-only Result output

✅ No contentious locks

- `PHYSICAL_CORES` may spin briefly on first access
- Atomic load only after init

✅ Tiny amount of overhead for chunking only

Don't just parallelize over scalar calcs!

Vectorize over chunks.

```rust
use rayon::prelude::*;
use std::sync::LazyLock;

/// Populated once on first access, then never again.
/// No lock used for access after initialization!
static PHYSICAL_CORES: LazyLock<usize> = LazyLock::new(num_cpus::get_physical);

/// Vector-parallel variant of [solid_angle_tetrahedron_scalar]
#[inline] // Enable cross-crate inlining
pub fn solid_angle_tetrahedra_par(
    tetrahedra: &[[[f64; 3]; 4]],
    out: &mut [f64],
) -> Result<(), &'static str> {
    // Chunk inputs. Only use real cores!
    let num_cores = rayon::current_num_threads().min(*PHYSICAL_CORES);
    let num_chunks = 1024.min(out.len() / num_cores);
    let (tet_chunks, out_chunks) = (
        tetrahedra.par_chunks(num_chunks),
        out.par_chunks_mut(num_chunks),
    );

    // Do vector calculations over each chunk in parallel
    (tet_chunks, out_chunks)
        .into_par_iter()
        .try_for_each(|(tetc, outc)| solid_angle_tetrahedron(tetc, outc))?;

    Ok(())
}
```

vector-parallel formula

vector formula

Eldritch horrors await

Thread Parallelism 3/3

- lscpu (native cpu info)
- pidstat (track thread preemption live)
- cachegrind (simulate cache misses)
- strace (track syscalls live)
- perf stat (estimate cache misses live)
- cargo-call-stack (inspect stack usage)
- stats_alloc (instrument global allocator)

```
~$ lscpu
...
Caches (sum of all):
  L1d:                   192 KiB (6 instances)
  L1i:                   192 KiB (6 instances)
  L2:                    3 MiB (6 instances)
  L3:                    32 MiB (2 instances)
```

```
~$ pidstat -w -p -t <pid>
12:01:58 AM UID   TGID TID cswch/s nvcswch/s Command
12:01:58 AM 1000 2983046 - 0.01 0.00 multi_daq
12:01:58 AM 1000 - 2983046 0.01 0.00 |__multi_daq
12:01:58 AM 1000 - 2984161 0.21 0.00 |__tsdb-dispatcher
12:01:58 AM 1000 - 2984162 0.70 0.01 |__csv-dispatcher
```

```
# Add these to configure a specific cache size:
# <size>,<associativity>,<line size>
# --I1=32768,8,64 \
# --D1=32768,8,64 \
# --LL=8388608,16,64 \
~$ valgrind --tool=cachegrind --cachegrind-out-file=./cachegrind.out <COMMAND>
~$ kcachegrind ./cachegrind.out # visualize
~$ cg_annotate --auto=yes ./cachegrind.out # show line numbers
~$ cg_annotate --source=yes ./cachegrind.out # annotate source code
```

```
~$ sudo perf stat -e cache-misses,cache-references,branches,branch-misses su - <USER> -c "<COMMAND>"
 Performance counter stats for 'su - <USER> -c <COMMAND>':
     262,051,652     cache-misses              #    6.88% of all cache refs
   3,807,478,401     cache-references
  93,183,164,713     branches
     923,190,394     branch-misses             #    0.99% of all branches
```

```
~$ strace -tt -f --syscall-times=ns -C <COMMAND>
# Track syscalls
# with forked threads (-f)
# with microsecond timestamps (-tt)
# with nanosecond-resolution durations (--syscall-times=ns)
# with a summary table at the end (-C)
```

Links

- InterpN @ v0.11.0 | github | crates.io | pypi | PGO workflow
- This slide deck | pitch.io | github | godbolt w/ FMA | godbolt w/o FMA
- The blog post | jlogan.dev

Further Reading

- Scalability! But at what COST?
- Jakub Beránek's blog post about cargo-pgo
- Nick Wilcox's auto-vectorizer tutorial
- Predrag Gruevski's blog post about optimizing cargo-semver-checks
- David Hewitt's talks about PyO3 (any of them, really)
- Miguel Raz's talk about SIMD for Sav-Gol filters
- Carl Kadie's blog post about guidelines for scientific libraries in Rust
- Ryan Levick's talk Rust before Main
- Herbie floating-point expression compiler

Acknowledgements

- Predrag Gruevski - mentor
- Clément Robert - contributor
- Yann Simon - reference CPUs
- Herold Aptroot - FMA latency
- Will Glyn - Herbie

Questions?

Concerns?

Pitch

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